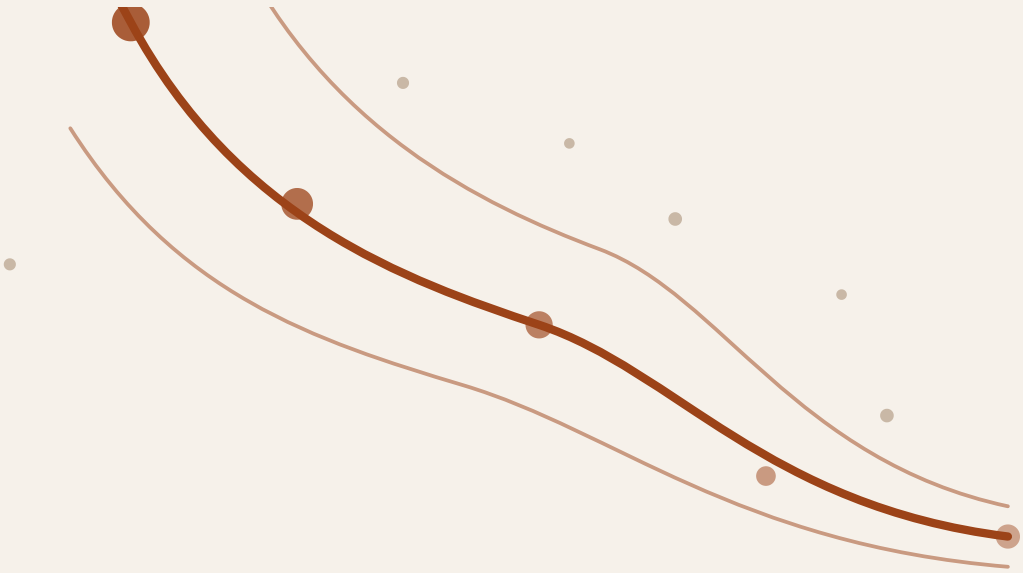
Innovation Management

From scattered initiatives
to a sustained organizational capability

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Chapter 10

10 Governance and portfolio management

In this chapter we address how innovation decisions are made: to whom each category of decision belongs, how the gates between stages are designed, how idea owners are treated fairly when the outcome is not what they hoped for, and why initiatives that should have been stopped keep going. We then turn to how initiatives are managed together as a balanced portfolio.

Introduction

Opening case: An innovation committee (composite case)

The innovation committee of one organization meets once every two months and is made up of eight senior managers. Each meeting's agenda carries between fifteen and twenty initiatives, with ten minutes allotted to each. An initiative is presented, a few general questions are asked, and the team is asked for "further study." Two years on, the portfolio held forty-two active initiatives; not one had been stopped, and not one had been scaled. Every team was working with a quarter of the resources it needed, waiting for the next meeting.

The committee was neither lenient nor strict; it was absent from the decision while present at the meeting. It never defined what it would decide itself and what it would leave to the teams, it never asked for different evidence at each stage, and it never stopped what needed stopping so as to free up resources for others. Closing these gaps is what the governance we set out in this chapter is built to do.

Learning Objectives

After studying this chapter, the reader should be able to:

١ Explain how innovation governance differs from the governance of stable operations.

٢ Allocate decision rights between governance levels and teams.

٣ Design gates whose criteria progress from assumptions to evidence.

٤ Apply the principles of procedural justice in dealing with idea owners.

٥ Explain escalation of commitment, and set kill criteria in advance.

٦ Manage initiatives as a balanced portfolio funded in stages.

10.1 Governance for uncertainty

Most governance tools in organizations were designed for stable work: an annual plan, a budget approved in advance, fixed performance indicators, and accountability for deviations from the plan. These tools work well when the path is known. In innovation, however, the path itself is unknown, and departing from the original plan may be evidence of learning. Innovation governance therefore fails in two directions: when the tools of stable work are applied to innovation, they stifle it; when innovation is exempted from governance altogether, it consumes resources without results.

Governance suited to innovation rests on a simple idea: investing in an initiative buys *learning* that reduces uncertainty, and every subsequent decision builds on what the team has learned. An initiative is not asked at the outset for what it cannot yet have, nor is it allowed at the end to continue without what it must have. This is what the following sections build.

10.2 Decision rights

Much governance confusion stems from ambiguous decision rights. Who decides whether an initiative continues? Who decides how it is carried out? Who decides how resources are allocated across initiatives? When these questions go unanswered, all decisions drift upward, and senior committees become crowded with details they lack the knowledge to judge, as in the opening case. Figure 10.1 proposes a distribution of decision rights across three levels that applies the principle of autonomy within boundaries: leadership determines the "what" and the "why," and teams own the "how."

Level	Decides	Does not decide
Senior leadership Board or steering committee, two or three times a year	Search fields and challenge questions, total investment, portfolio balance, and the big bets	The details of individual initiatives and how they are carried out
Portfolio committee Monthly or at gates	Gate decisions: continue, modify, merge, suspend, or stop, and funding for the next stage	How the team works within a stage
The team and its sponsor Continuously	Designing experiments, ways of working, and use of the approved stage resources	Exceeding the stage's limits or resources without returning to the gate

Figure 10.1 The distribution of decision rights in innovation governance. Prepared by the author.

In this distribution, senior leadership's role remains the most far-reaching even though it makes fewer decisions: it determines where the organization searches, how much risk it takes, and how it balances the near term against the long term. These decisions are often neglected because they are harder than reviewing individual initiatives. The international standard for innovation management systems assigns top management responsibility for setting innovation policy and strategy, providing resources, and defining roles and authorities (ISO, 2024), which is consistent with this distribution.

Decision rights are best written down and announced, so that the team knows what it can decide on its own without waiting, and when it must return to the gate. Putting them in writing protects both sides: it shields the team from interference in the details of its work, and it shields the organization from initiatives expanding without a decision.

10.3 Gates: from assumptions to evidence

Among the most widespread tools of innovation governance is the **Stage-Gate** system developed by Cooper. The system divides an initiative's journey into stages separated by gates, at which a decision is made to continue, stop, suspend, or recycle the work according to criteria set in advance (Cooper, 1990). The value of a gate is that it turns a decision from a general impression into a specific question: has the initiative generated enough knowledge to merit investment in the next stage?

The most important principle in gate design is that **the criteria change from one stage to the next**. At the early gates, it is enough that the problem matters and the assumptions are stated. With each gate, stronger evidence is required, until the initiative reaches the scaling gate with evidence of adoption, cost, and sustainability. Table 3.1 set out the leader's question and the type of evidence at each stage, and Figure 10.2 illustrates the logic of this progression.

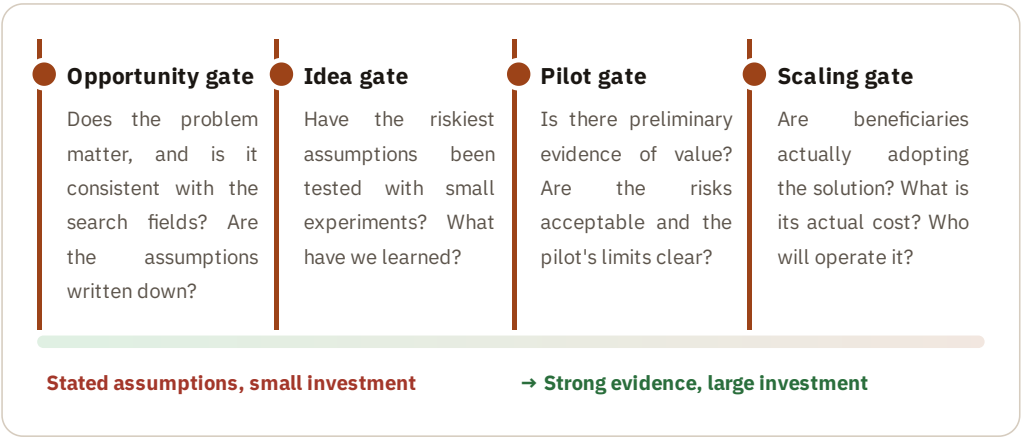


Figure 10.2 Gate criteria progressing from assumptions to evidence, with investment increasing alongside them. Prepared by the author, drawing on Cooper (1990).

The Stage-Gate system has been criticized for becoming slow and bureaucratic, and for fitting poorly with highly novel initiatives in which the understanding of the problem itself shifts as the work proceeds. Cooper responded to this criticism with hybrid models that combine gates with agile ways of working: shorter stages, lighter gates for small initiatives, and short build-and-test cycles within each stage (Cooper, 2014). The lesson is that a gate is first and foremost a tool for learning and decision making, and that its depth should match the size of the risk: a small, low-cost experiment does not need the same gate as rolling out a solution across the entire organization, as Chapter 3 cautioned.

Caution

A gate that always ends in approval is not a gate. If no initiative has been stopped at a gate over a full year, then either the organization chooses its ideas with rare brilliance, or the gates are not working.

10.4 Procedural justice

Chapter 5 presented three ways to prevent decision limbo: an announced response time, a reasoned response, and periodic review. One question remains: why does a reasoned response matter so much to the idea owner, even when it is a rejection? Research on **procedural justice** provides the answer. Kim and Mauborgne showed that people accept decisions that do not go their way, and continue to cooperate, if they see the process by which those decisions were made as fair. This fairness rests on three principles: *engagement*, giving those concerned the opportunity to present their views and ideas; *explanation*, setting out the reasons for the decision; and *expectation clarity*, making known what will be required next and the criteria by which it will be judged (Kim & Mauborgne, 1997).

In innovation governance, these principles translate into specific practices:

- **Criteria announced before evaluation:** idea owners know in advance on what basis they will be evaluated, so they are not surprised by a criterion that was never mentioned.
- **An opportunity to present in person:** at major gates, the team presents its initiative itself and answers questions, rather than having it discussed in its absence.
- **A response with four elements:** what was decided, why, what would have changed the decision, and what next step is open to the idea owner.
- **Attributing the idea to its owner:** the idea owner's name stays attached to the idea through every stage, and they are involved in implementing it if they wish, rather than having it handed to another unit that carries it out without them.

In this way, governance itself becomes a means of meeting the innovator's needs set out in Chapter 7, particularly recognition and understanding. An organization that rejects an idea fairly keeps its owner; one that accepts an idea without crediting its owner may lose them.

10.5 Why don't initiatives stop?

Innovation portfolios suffer from a problem that is the reverse of decision limbo: initiatives that continue even though the evidence does not support them. Staw studied this phenomenon, which he called **escalation of commitment**. In his experiment, participants who were personally responsible for a failing investment decision allocated more additional funds to it than did those who were not responsible for it (Staw, 1976). People tend to persist with a decision they have made in order to justify their original choice, even when new evidence points the other way.

In innovation governance, escalation takes familiar forms: an initiative sponsored by a senior leader, which no one dares propose stopping; an initiative on which a great deal has been spent, prompting the refrain "we can't stop now"; and an initiative run by an enthusiastic team that, it is feared, would be demoralized by a stop. Three practices curb this escalation:

- **Kill criteria set in advance:** at the start of each stage, the results that would stop the initiative, whether achieved or missed, are defined before anyone becomes attached to it.
- **Separating the decision from its original owner:** parties other than the sponsor who championed the initiative take part in the decision to continue.
- **Honoring a good stop:** a team that stopped its initiative at the right time and documented what it learned has done good work, and is treated accordingly. This is a direct application of the concept of intelligent failure from Chapter 8.

This explains the opening case: a committee that stops nothing spreads innovation resources across more initiatives than it can fund, and so weakens them all.

10.6 Portfolio management

If gates govern the individual initiative, portfolio management governs initiatives collectively. In their study of company practices, Cooper, Edgett, and Kleinschmidt found that portfolio management pursues three goals: maximizing total value, balancing types of initiatives, and aligning with strategy. They also found that one of the most common problems is having more initiatives than resources allow, so that all of them are delayed (Cooper et al., 2001). Figure 10.3 illustrates these goals.

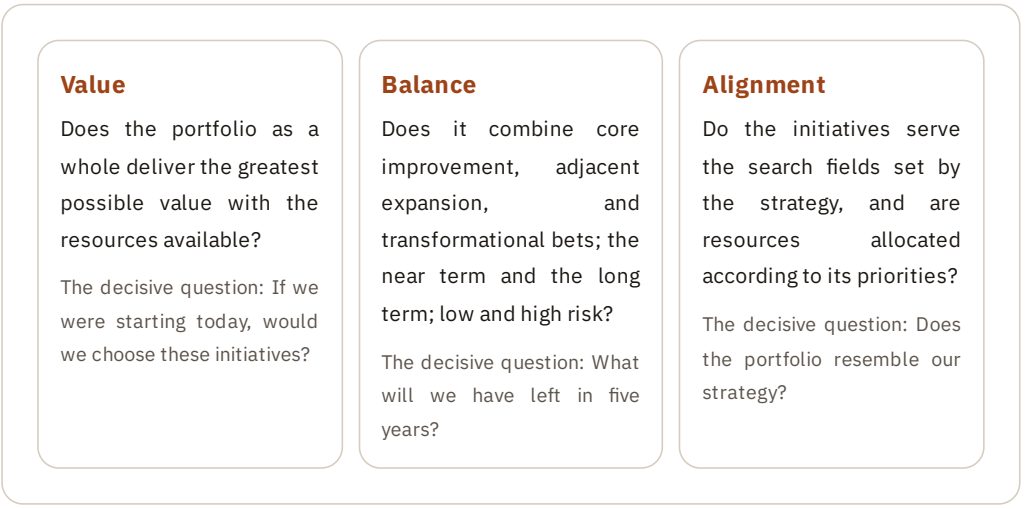


Figure 10.3 The goals of innovation portfolio management, adapted from Cooper et al. (2001).

Balance is tied to the degrees of innovation presented in Chapter 1: incremental, adjacent, and radical. Nagji and Tuff propose that resources be distributed among them in proportions suited to the nature of the organization (Nagji & Tuff, 2012). The right proportion varies from one organization to another, but a portfolio that lacks any one of the three types calls for an explicit question about why.

One of the most important tools of portfolio management is **staged funding**. Rather than approving an initiative's full budget at the outset, it is funded stage by stage, with each stage's funding tied to passing its gate. Gompers showed that venture capital firms rely on this approach because it preserves their

ability to review and stop, and that they shorten the interval between funding rounds as uncertainty increases (Gompers, 1995). The logic of **real options** explains why: a small investment at an early stage is like buying an option that allows the organization to invest more if the opportunity materializes, or to stop with a limited loss if it does not (McGrath, 1999). By this logic, early failure becomes a reasonable price for knowledge that cannot be gained without experimentation.

The portfolio concept extends beyond the initiatives within a single unit. In organizations with multiple innovation units, such as the university described in Chapter 6 with its accelerator, its digital fabrication lab (Fab Lab), and its centers, each unit manages its own portfolio in isolation, so initiatives are duplicated and ideas get lost in the gaps between units. The alternative is a shared portfolio: a single entry point for ideas, shared gates, and the movement of an initiative from one unit to another according to its stage. We return to this design in Chapter 20.

In Practice

List all the active initiatives in your organization, then ask of each: If it were presented to us today for the first time, knowing what we know now, would we start it? The initiatives for which the answer is “no” are candidates for stopping, and their resources are candidates for initiatives that deserve them more.

Chapter tool: The gate charter

For each gate in your system, write a short charter that answers the following questions, and share it with teams before the stage begins. If you cannot answer one of the questions, that is a governance gap that should be closed before the stage begins.

Chapter tool The gate charter (example: the pilot gate)

Item	Question	Example
Gate question	What do we need to know in order to invest in the next stage?	Is there preliminary evidence that the solution addresses the problem for beneficiaries?
Evidence required	What exactly does the team submit?	The results of at least two small experiments, and what the team learned from them
Decision owner	Who decides? Who is consulted?	The portfolio committee, after hearing from the team and the sponsor
Response time	When is the decision issued once the submission is complete?	Within two weeks
Possible decisions	What options are available?	Continue, modify, merge, suspend, or stop
Kill criteria	What stops the initiative? (Defined at the start of the stage)	More than half of the beneficiaries in the pilot do not use the solution
Funding for the next stage	What resources does passing the gate unlock?	The pilot budget and the time of two dedicated members
Informing the team	How is the decision communicated?	A written response containing the four elements, and a discussion with the team

Chapter Summary

- Innovation governance differs from the governance of stable work: investment buys learning that reduces uncertainty, and every decision builds on what the team has learned.
- Decision rights are distributed across three levels: senior leadership for direction and balance, the portfolio committee for gates, and the team for ways of working.
- Gate criteria progress from assumptions to evidence, and their depth is proportionate to the size of the risk.
- People accept decisions that go against their interests when they are confident the process is fair: through engagement, explanation, and clear expectations.
- Escalation of commitment keeps alive initiatives that should have been stopped; it is curbed by setting kill criteria in advance, separating the decision from its original owner, and honoring a good stop.
- Portfolio management seeks value, balance, and alignment, and avoids piling up initiatives on limited resources.
- Staged funding preserves the ability to review, and makes early failure a limited price for knowledge.

Key Terms

Decision rights Decision rights	A specification of who has the authority to make each type of innovation decision.
Stage-Gate Stage-Gate	A system that divides an initiative's journey into stages separated by decision points with criteria set in advance.
Procedural justice Procedural justice	The fairness of the way a decision is made, based on engagement, explanation, and expectation clarity.
Escalation of commitment Escalation of commitment	Continuing to invest in a failing course of action in order to justify the original decision.
Staged funding Staged funding	Funding an initiative stage by stage, with each tranche tied to passing a gate.
Real options Real options	A logic that views a small early investment as a right to expand later or to stop with a limited loss.

Review and Discussion Questions

1. Why do governance tools designed for stable work fail when applied to innovation? Understanding
2. Analyze the committee in the opening case using the concepts of decision rights, gates, and escalation of commitment, and propose three changes. Analysis

1. Write a charter for one gate in your organization's innovation system, including kill criteria. **Application**

2. Write a response to the owner of an idea that was not accepted, including the four elements of a fair response. **Application**

3. "Stopping an initiative after a great deal has been spent on it is a waste of public money." Evaluate this statement in light of the logic of real options. **Evaluation**

Chapter references

Cooper, R. G. (1990). Stage-gate systems: A new tool for managing new products. *Business Horizons, 33*(3), 44–54. https://doi.org/10.1016/0007-6813(90)90040-I

Cooper, R. G. (2014). What's next? After stage-gate. *Research-Technology Management, 57*(1), 20–31. https://doi.org/10.5437/08956308X5606963

Cooper, R. G., Edgett, S. J., & Kleinschmidt, E. J. (2001). Portfolio management for new product development: Results of an industry practices study. *R&D Management, 31*(4), 361–380. https://doi.org/10.1111/1467-9310.00225

Gompers, P. A. (1995). Optimal investment, monitoring, and the staging of venture capital. *The Journal of Finance, 50*(5), 1461–1489. https://doi.org/10.1111/j.1540-6261.1995.tb05185.x

International Organization for Standardization. (2024). *Innovation management system — Requirements* (ISO Standard No. 56001:2024). https://www.iso.org/standard/79278.html

Kim, W. C., & Mauborgne, R. (1997). Fair process: Managing in the knowledge economy. *Harvard Business Review, 75*(4), 65–75.

McGrath, R. G. (1999). Falling forward: Real options reasoning and entrepreneurial failure. *Academy of Management Review, 24*(1), 13–30. https://doi.org/10.5465/amr.1999.1580438

Nagji, B., & Tuff, G. (2012). Managing your innovation portfolio. *Harvard Business Review, 90*(5), 66–74.

Staw, B. M. (1976). Knee-deep in the big muddy: A study of escalating commitment to a chosen course of action. *Organizational Behavior and Human Performance, 16*(1), 27–44. https://doi.org/10.1016/0030-5073(76)90005-2

How does an organization move from scattered initiatives that celebrate innovation to a sustained capability to produce it?

This book traces the path of innovation from the first signal to a solution adopted in practice that creates value, and analyzes the organizational foundations that make this path a settled practice: strategy, systems, culture, governance, teams and measurement, with people at the center.

It rests on three pillars: a clear, high-priority problem; motivated people whose needs are met; and a system that connects them and turns effort into repeatable results. It combines international research evidence with the Saudi context, and pairs each chapter with a tool that can be used in a real organization.

Its twenty chapters are organized in five parts: foundations; institutional foundations and people; the initiative journey; value and the ecosystem; and an application to the university innovation ecosystem.